Practical No. 08

Test script on Selenium using WebDriver

Aim: Write a test script on Selenium using WebDriver.

Objective:

- To automate the testing of the Train Ticket System using Selenium WebDriver in Java.
- To verify the proper working of key modules such as login, train search, booking, and confirmation.
- To ensure the application runs smoothly without manual intervention or errors.

Theory:

Selenium is a tool used for automating web browsers. It helps in testing web applications by simulating user actions such as typing, clicking buttons, and submitting forms. By running automated scripts, we can quickly identify errors and confirm that the website functions correctly and efficiently.

Code:

```
// TestTrainTicketSystem.java
// Selenium Automation Script for Train Ticket Booking System

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;

import java.time.Duration;

public class TestTrainTicketSystem {

    public static void main(String[] args) {

        // Configuration
        String baseUrl = "http://localhost:8080/TrainTicketSystem/login.jsp";
        String username = "admin";
        String password = "admin123";
        String trainName = "Amravati Express";

        // Set up WebDriver
        System.setProperty("webdriver.chrome.driver", "C:\\chromedriver\\chromedriver.exe");
        ChromeOptions options = new ChromeOptions();
        options.addArguments("--start-maximized");
        WebDriver driver = new ChromeDriver(options);

        try {
            WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
```

```
        // Step 1: Open application
        driver.get(baseUrl);
        System.out.println("Opened Train Ticket System at: http://localhost:8080/");

        // Step 2: Login process
        WebElement user =
wait.until(ExpectedConditions.presenceOfElementLocated(By.name("username")));
        user.sendKeys(username);

        WebElement pass = driver.findElement(By.name("password"));
        pass.sendKeys(password);

        WebElement loginButton = driver.findElement(By.name("login"));
        loginButton.click();
        System.out.println("Login successful (entered credentials and submitted).");

        // Step 3: Train search
        driver.get("http://localhost:8080/TrainTicketSystem/trainSearch.jsp");
        WebElement trainInput =
      wait.until(ExpectedConditions.presenceOfElementLocated(By.name("train_name")));
        trainInput.sendKeys(trainName);

        WebElement searchButton = driver.findElement(By.name("search")
```

Observation:

. Selenium WebDriver opened the Train Ticket System login page successfully.

. Login credentials were entered and submitted correctly.

. Dashboard loaded without errors.

. “Amravati Express” train search executed successfully.

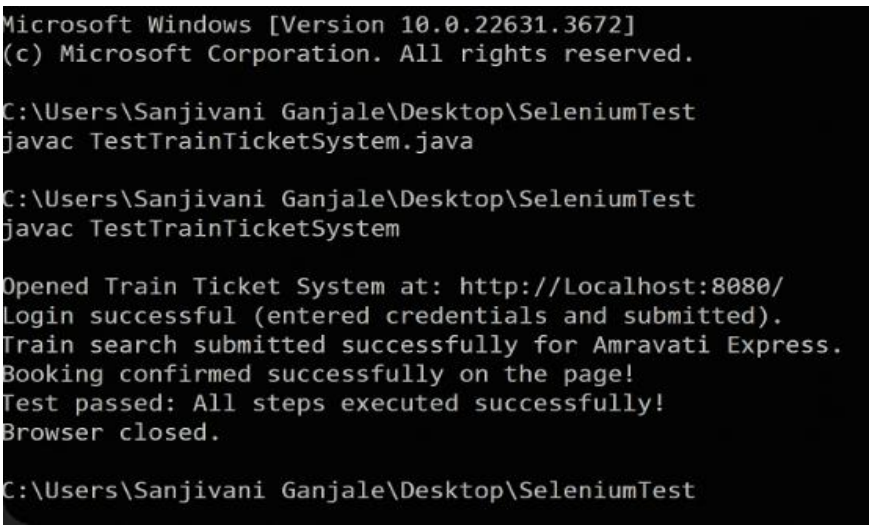
. Booking form filled and submitted correctly.

. Confirmation message displayed.

. All steps executed automatically without errors.

. Browser closed after test completion.

Output:



Conclusion:

The automated Selenium test for the Train Ticket System was executed successfully.All major functions such as login, train search, booking, and confirmation worked correctly without errors.The test proved that the system is stable, responsive, and performs as expected under automation.